

Lecture 3: Block Diagram of a Computer

Input Unit · Output Unit · CPU (Control Unit, ALU, Registers) · Memory Unit · System Bus

Learning Outcomes

- Define a block diagram, and explain why it is useful for understanding how a computer is organised.
- Draw and label a complete, detailed block diagram of a computer.
- Explain the function of the Input Unit and list at least five common input devices with examples.
- Explain the function of the Output Unit and list at least four common output devices, distinguishing soft copy from hard copy.
- Describe the CPU's three internal components — the Control Unit, the ALU, and Registers — and explain the Fetch → Decode → Execute cycle.
- Differentiate between Primary and Secondary memory in the context of the block diagram.
- Explain the System Bus and its three types: Address Bus, Data Bus, and Control Bus.
- Trace the complete flow of data and control signals through the block diagram for a simple instruction.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations, diagrams, and examples.

1 What Is a Block Diagram, and Why Do We Need One?

In Lecture 1, a computer was defined as an electronic device that works on a simple Input → Process → Output cycle, and we saw a first, simplified look at its basic parts. This lecture opens up that idea fully.

A **block diagram** is a simplified diagram that represents the major functional units of a system as labelled boxes, connected by lines or arrows that show how data and control signals move between them. It deliberately leaves out low-level detail — individual wires, transistors, or chip pin-outs — and instead shows only the logical organisation: what the major parts are, and how they relate to one another.

2 The Five Basic Units — Overview

Every computer, regardless of its size, brand, or purpose, is built from the same five basic functional units.

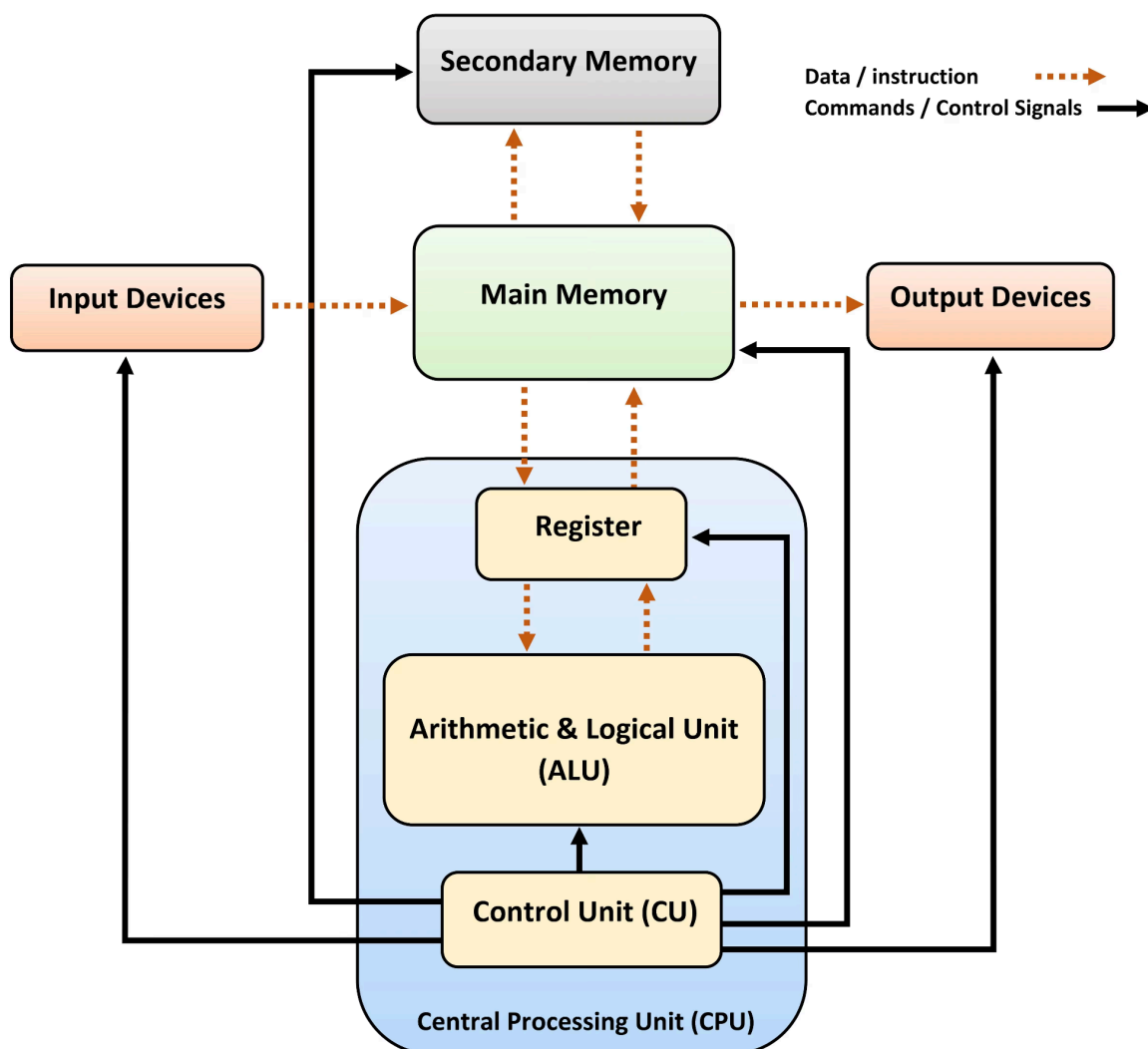


Fig 3.1 — The complete block diagram of a computer.

Unit	One-line Function
Input Unit	Accepts data and instructions from the outside world and converts them into binary form.
Central Processing Unit (CPU)	Processes data — performs calculations and comparisons, and directs every other unit. Made up of the Control Unit, the ALU, and Registers.
Output Unit	Converts the CPU's binary result back into a form a human being can understand.
Memory Unit	Stores data, instructions, and intermediate results — split into Primary Memory and Secondary Memory.
System Bus	The set of shared pathways that physically connects all of the above units, carrying addresses, data, and control signals between them.
Important Point for Exams The Control Unit, the ALU, and a small set of Registers are together called the Central Processing Unit (CPU) — the “brain” of the computer. Memory, Input, and Output are separate units that the CPU works with, not physically part of the CPU itself. Examiners frequently test exactly this boundary: what is “inside” the CPU, and what is merely “connected to” it.	

3 The Input Unit — In Detail

The Input Unit is the part of a computer system that accepts data and instructions from the outside world — usually from a human being — and converts them into a machine-readable, binary form that the CPU can process.

How the Input Unit Works

1. It accepts raw data and instructions from the user or from another device.
2. It converts this data into binary digits (0s and 1s) — the only language the CPU understands.
3. It supplies the converted data to the memory unit, or directly to the CPU along the system bus, for processing.

Common Input Devices

Device	What It Does
Keyboard	Converts key presses into character codes.
Mouse	Converts hand movement and clicks into cursor position and selection commands.
Scanner	Converts a printed image or document into a digital image.
Joystick	Converts hand movement into direction and control signals, mainly for games and simulations.

Device	What It Does
Microphone	Converts sound waves into digital audio data.
Light Pen / Touch Screen	Converts the position touched or pointed at on a screen into coordinate data.
MICR (Magnetic Ink Character Reader)	Reads characters printed with magnetic ink, widely used to process cheques in banks.
OMR (Optical Mark Reader)	Detects the presence of marks (such as filled bubbles) on a printed sheet, widely used to check answer sheets.
Barcode / QR Scanner	Converts a printed code into a product or payment ID.
Important Point for Exams Memorise this one-sentence definition: the Input Unit converts human-understandable data into machine-understandable (binary) data. Examiners frequently ask you to “name any four or five input devices” — always add one real-life example or use next to each device for full marks.	

4 The Output Unit — In Detail

The Output Unit is the part of a computer system that takes the processed result from the CPU — which exists internally as binary data — and converts it into a form a human being can understand and use, such as text, images, sound, or a printed page. It performs exactly the reverse job of the Input Unit.

Common Output Devices

Device	What It Produces
Monitor (VDU)	A visual display on screen — a soft copy.
Printer	A printed page — a hard copy.
Speaker	Sound.
Plotter	A large-format printed drawing, such as an engineering site map.
Projector	A large projected image, such as a lecture slide on a classroom screen.

Device	What It Produces
Soft Copy vs Hard Copy A soft copy is a temporary, electronic form of output — such as text on a monitor — that disappears when the device is switched off. A hard copy is a permanent, physical form of output — such as a printed page — that continues to exist once the computer is switched off. This distinction is a frequently asked one-mark or short-answer question.	

Comparison: Input Unit vs Output Unit

Point	Input Unit	Output Unit
Direction of conversion	Human-understandable form → binary	Binary → human-understandable form
Position in the cycle	Performs the “Input” stage	Performs the “Output” stage
Data flow	Data flows INTO the CPU/memory	Data flows OUT of the CPU/memory
Examples	Keyboard, Mouse, Scanner, Microphone	Monitor, Printer, Speaker, Plotter

5 The Central Processing Unit (CPU)

The Central Processing Unit (CPU) is the most important component of a computer system — often called its “brain” — because it is responsible for fetching, interpreting, and executing every instruction in a program, and for coordinating the activity of every other unit. The CPU is not one single block; it is built from three smaller parts working together.

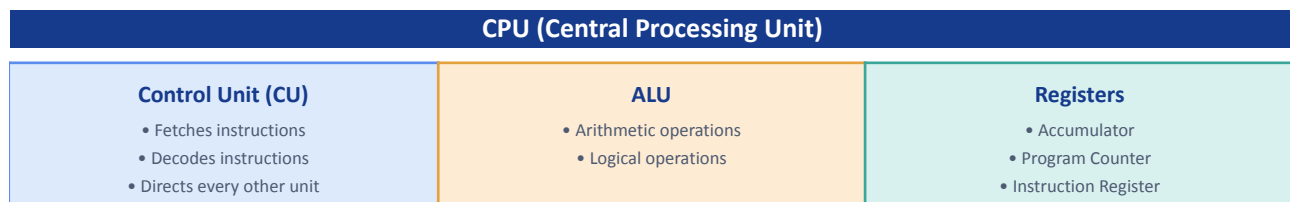


Fig 3.2 — Inside the CPU: the Control Unit, the ALU, and Registers.

5.1 The Control Unit (CU)

The Control Unit performs no calculations itself. Instead, it directs and coordinates the activities of every other part of the computer, by generating precisely timed electronic control signals that tell each component exactly when to act. As

the name suggests, this unit controls all the functions of the computer, and every other device or part of the computer interacts through the Control Unit.

The Control Unit repeatedly performs a three-step cycle for every single instruction in a program:

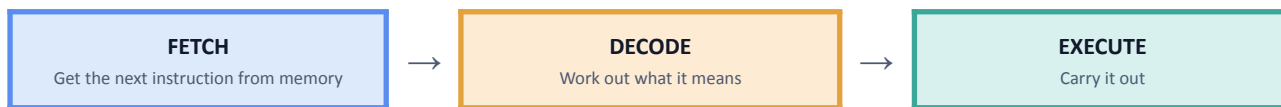


Fig 3.3 — The Fetch → Decode → Execute cycle, repeated continuously for every instruction.

Common Doubt — Does the Control Unit Calculate?

This is a favourite examiner's trick question. The Control Unit never performs a calculation itself — it only reads out the next instruction and tells the correct unit to act on it. All arithmetic and logical calculation is done by the ALU. Confusing the two is one of the most common mistakes students make in this topic.

5.2 The Arithmetic Logic Unit (ALU)

The Arithmetic Logic Unit is the part of the CPU where all arithmetic operations (addition, subtraction, multiplication, division) and logical operations (comparisons such as greater-than, less-than, and equal-to; and logical operations such as AND, OR, NOT) actually take place.

Type	Examples of Operations	What It Is Used For
Arithmetic operations	Addition (+), Subtraction (−), Multiplication (×), Division (÷)	Calculating totals, interest, averages, bills
Logical operations	AND, OR, NOT, and comparisons (>, <, =)	Making decisions and comparing two values
Real-Life Example When a bank's computer checks whether a customer's account balance (₹5,000) is greater than or equal to the withdrawal amount requested at an ATM (₹2,000) before dispensing cash, this comparison is a logical operation carried out by the ALU — not a decision made by the machine “thinking” about it.		

Comparison: ALU vs Control Unit

Point	ALU	Control Unit
Main function	Performs arithmetic and logical operations	Directs and coordinates all other components
Does it calculate?	Yes — this is its entire job	No — it never calculates
Does it generate control signals?	No	Yes — this is its entire job

Point	ALU	Control Unit
6 The Memory Unit		

The Memory Unit stores all input data, instructions, and any data produced part-way through processing, so that the CPU can access it as needed. Memory is of two types: Primary Memory and Secondary Memory.

Point	Primary Memory	Secondary Memory
Position	Directly and closely coupled to the CPU for fast access	External to the CPU
Speed	Very fast	Much slower than primary memory
Capacity	Smaller, measured in gigabytes	Much larger, measured in gigabytes to terabytes
Volatility	RAM is volatile (loses data when powered off); ROM is non-volatile	Non-volatile — keeps data even without power
Examples	RAM, ROM, Cache	Hard Disk (HDD), SSD, USB drive
Cost per byte	Higher	Lower

Important Point for Exams

Many textbooks loosely say primary memory “resides within” the CPU. Strictly speaking, only registers and cache are physically built into the CPU chip itself; RAM and ROM are separate chips placed very close to the CPU for fast access. For exam purposes, both phrasings are accepted, but you should be able to explain the distinction if asked to elaborate — exactly as covered in Lecture 1.

7 The System Bus — Connecting Everything

A **bus** is a set of shared electronic pathways (wires) used to transfer data, addresses, or control signals between the components of a computer. Every unit in the block diagram — the CPU, memory, input devices, and output devices — is physically connected together by the system bus.

SYSTEM BUS		
Address Bus • Carries WHERE: the memory address • One-directional (CPU → memory/I-O)	Data Bus • Carries WHAT: the actual data • Bidirectional (both ways)	Control Bus • Carries HOW/WHEN: Read, Write, Interrupt signals • Coordinates timing

Fig 3.4 — The three types of system bus, and what each one carries.

Bus Type	Direction	Carries	Simple Analogy
Address Bus	One-directional (CPU to memory/I-O)	The memory address to be accessed	The house number on an envelope
Data Bus	Bidirectional (data flows both ways)	The actual data being read or written	The letter inside the envelope
Control Bus	Carries signals in both directions as needed	Read/Write signals, clock timing, interrupts	The postal instructions: “deliver” or “collect,” and when
Memory Trick for Exams Address = WHERE, Data = WHAT, Control = WHEN and HOW. If a question asks you to name the three buses, always pair each one with what it carries — that pairing is usually worth its own mark.			

8 Complete Data and Control Flow — A Walkthrough

The block diagram is easiest to understand by tracing exactly what happens, step by step, when a computer carries out one simple instruction — for example, adding two numbers that a user has typed in.

1. The user types 5 + 3 on the keyboard. The Input Unit converts each key press into binary and passes it toward memory.
2. The binary data and the program's instructions are held in the Memory Unit until the CPU needs them.
3. The Control Unit's Program Counter (PC) holds the address of the next instruction; that address is placed on the Address Bus and sent to memory.
4. Memory responds by placing the requested instruction on the Data Bus; it travels back and is loaded into the Instruction Register (IR) — this is the Fetch step.
5. The Control Unit decodes the instruction in the IR, working out that it is an addition instruction, and identifies that the ALU must be involved — this is the Decode step.
6. The Control Unit sends control signals (along the Control Bus) directing the ALU to add the two values; the ALU performs the addition and the result is placed in the Accumulator — this is the Execute step.
7. The Control Unit sends the result, via the bus, to the Output Unit, which converts it from binary back into a human-readable form.
8. The monitor displays the number 8 on screen.

Important Point for Exams

This single walkthrough uses every unit in the block diagram and all three buses. If a long-answer question asks you to “explain how a computer executes an instruction” or “trace the flow of data,” structure your answer exactly like this — as a numbered sequence naming the unit and the bus involved at each step.

9 Putting It All Together — A Worked Example

Consider a shopkeeper using a computerised billing system to calculate the total price of items purchased:

Step	Unit Involved	What Happens
1	Input Unit	The shopkeeper scans each item's barcode using a barcode scanner (data in).
2	Memory Unit	The price of each scanned item is looked up and temporarily held in memory.
3	Control Unit	Fetches and decodes the “add item price to running total” instruction for each item scanned.
4	ALU	Adds each item's price to the running total — an arithmetic operation.
5	System Bus	Carries the item's address, its price data, and the control signals between memory, the CPU, and the output device throughout this process.
6	Output Unit	The monitor displays the running total, and the printer produces a final printed bill — a hard copy — once billing is complete.

This everyday example uses every unit from this lecture: the Input Unit, the Memory Unit, all three parts of the CPU, the System Bus connecting them, and the Output Unit — exactly the organisation shown in the block diagram.

10 Fill in the Blanks

- 1. A _____ is a simplified diagram showing the major functional units of a computer and how they are connected.
- 2. The unit that accepts data and instructions from the outside world is called the _____.
- 3. The unit that presents the processed result to the user is called the _____.
- 4. The _____ directs and coordinates every other unit of the computer, but never performs calculations itself.
- 5. The _____ performs all arithmetic and logical operations inside the CPU.
- 6. The register that holds the address of the next instruction to be fetched is the _____.
- 7. _____ memory is closely coupled to the CPU for fast access, while _____ memory is external to it.
- 8. The bus that carries the actual data being transferred is called the _____ bus.
- 9. The bus that carries the memory address to be accessed is called the _____ bus.
- 10. The three-step cycle the Control Unit repeats for every instruction is Fetch, _____, and Execute.

Answers

1. Block diagram | 2. Input Unit | 3. Output Unit | 4. Control Unit | 5. ALU (Arithmetic Logic Unit) | 6. Program Counter (PC) | 7. Primary, Secondary | 8. Data | 9. Address | 10. Decode

11 True or False

- 1. The Control Unit performs all arithmetic calculations in a computer.
- 2. The Input Unit converts binary data into human-readable form.
- 3. The ALU performs both arithmetic and logical operations.
- 4. Registers are located inside the CPU.
- 5. Secondary memory is generally faster than primary memory.
- 6. The address bus is used to carry the actual data being transferred.
- 7. A monitor is an example of a soft-copy output device, while a printer produces a hard copy.
- 8. The system bus connects the CPU to memory and input/output devices.

Answers with Explanation

- 1. False.** The Control Unit never calculates — that is the ALU's job. The Control Unit only directs and coordinates.
- 2. False.** It is the reverse: the Input Unit converts human-understandable data into binary; the Output Unit converts binary back into human-readable form.
- 3. True.** Arithmetic operations (like addition) and logical operations (like comparisons) are both carried out by the ALU.
- 4. True.** Registers are tiny, extremely fast storage locations built directly inside the CPU.
- 5. False.** It is the reverse: primary memory is much faster than secondary memory, though secondary memory has far greater capacity.
- 6. False.** The address bus carries the memory address to be accessed; the data bus carries the actual data.
- 7. True.** A monitor displays a soft copy (temporary, on-screen); a printer produces a hard copy (permanent, on paper).
- 8. True.** The system bus is precisely the shared set of pathways that links the CPU, memory, and I/O devices together.

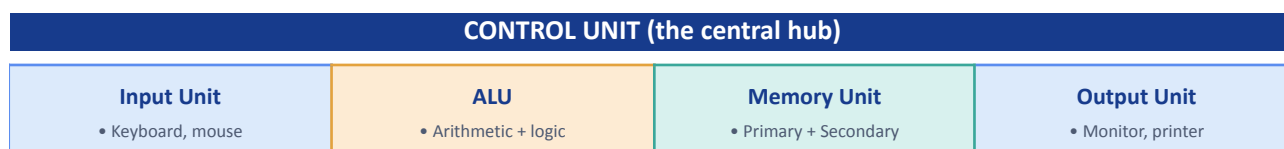
12 Short Answer Questions (2–3 Marks)

- What is a block diagram? Why is it useful? (2 marks)
- List the five basic units of a computer. (2 marks)
- Differentiate between the Input Unit and the Output Unit. (3 marks)
- What is the function of the Control Unit? (2 marks)
- Differentiate between the ALU and the Control Unit. (3 marks)
- What is a register? Name any two registers with their function. (3 marks)
- Differentiate between Primary and Secondary memory. (3 marks)
- What are the three types of buses in a computer, and what does each carry? (3 marks)
- Explain the Fetch → Decode → Execute cycle in brief. (3 marks)

13 Long Answer Model Answers (5, 10 & 15 Marks)

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

Q1 (5 Marks): Draw and explain the basic block diagram of a computer.



(A complete labelled block diagram, as in Fig 3.1 of these notes, should be drawn here in your answer sheet to earn full diagram marks.)

Every computer is built from five basic units. The Input Unit (keyboard, mouse, scanner) brings data and instructions into the system, converting them into binary form. The Control Unit directs and coordinates every other part of the computer, acting as the central hub through which all communication passes. The Arithmetic Logic Unit carries out every arithmetic and logical operation. The Memory Unit stores data, instructions, and intermediate results, split into Primary Memory (fast, closely coupled to the CPU) and Secondary Memory (external, larger, and permanent). Finally, the Output Unit (monitor, printer) presents the processed result back to the user, converting it from binary into a human-readable form. The Control Unit and ALU together, with a small set of registers, form the Central Processing Unit (CPU) — the “brain” of the computer.

Q2 (10 Marks): Explain the CPU with reference to the Control Unit, the ALU, and Registers, and describe the Fetch → Decode → Execute cycle.

The Central Processing Unit (CPU) is the most important component of a computer system, often called its “brain,” because it is responsible for fetching, interpreting, and executing every instruction in a program. The CPU itself is made up of three smaller parts working together: the Control Unit, the Arithmetic Logic Unit (ALU), and a small set of Registers.

Part	Function
Control Unit (CU)	Directs and coordinates every other component by generating precisely timed control signals; performs no calculations itself.



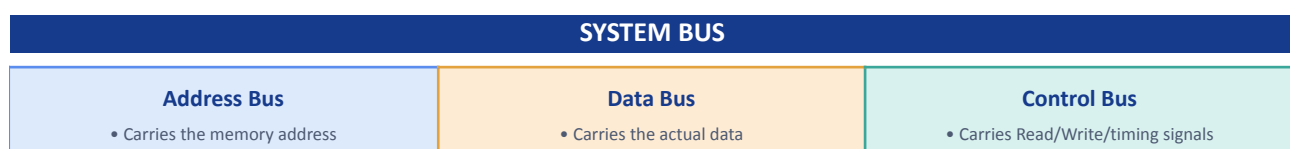
(The Fetch → Decode → Execute cycle, as in Fig 3.3 of these notes.)

The Control Unit repeatedly performs a three-step cycle for every instruction: in the Fetch step, the next instruction's address (held in the Program Counter) is used to retrieve that instruction from memory into the Instruction Register; in the Decode step, the Control Unit works out what the instruction means and which other parts of the CPU need to be involved; and in the Execute step, the instruction is carried out, often by directing the ALU to perform a calculation, with the result placed in the Accumulator. This cycle repeats continuously, billions of times per second, for as long as the computer is switched on.

In conclusion, no single part of the CPU could do this job alone: the Control Unit ensures every instruction is fetched and interpreted correctly, the ALU performs the actual calculation or comparison, and the Registers provide the small, fast working space both of them depend on.

Q3 (10 Marks): Explain the System Bus and its three types, with a diagram.

Every unit in a computer's block diagram — the CPU, memory, and input/output devices — must be physically connected so that data, addresses, and timing signals can travel between them. This shared set of electronic pathways is called the system bus, and it is divided into three types, each carrying a different kind of information.



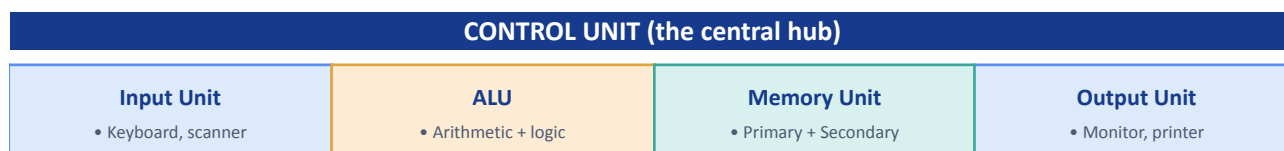
(The three types of system bus, as in Fig 3.4 of these notes.)

The Address Bus carries the memory address the CPU wants to access — it tells the rest of the system “where” to read from or write to, and normally carries signals only from the CPU outward. **The Data Bus** carries the actual data being transferred — the “what” — and is bidirectional, since data must be able to flow both into and out of the CPU. **The Control Bus** carries control signals such as Read, Write, and interrupt requests, along with timing information, coordinating exactly when each transfer should happen.

In conclusion, the address, data, and control buses work together as a coordinated team on every single memory access: the address bus says where, the data bus carries what, and the control bus decides how and when — together forming the physical backbone that connects every unit shown in the block diagram.

Q4 (15 Marks): “A computer's block diagram shows organisation, not just parts.” Discuss this statement by explaining how the Input Unit, CPU, Memory Unit, Output Unit, and System Bus work together to execute a single instruction, using a suitable example.

This statement is accurate. A block diagram is valuable not simply because it lists five separate boxes, but because it shows exactly how those boxes cooperate, in a fixed sequence, to carry out even the simplest task.



(A complete labelled block diagram, as in Fig 3.1 of these notes, should be drawn here in your answer sheet to earn full diagram marks.)

The Input Unit begins the process, converting a human action — such as typing on a keyboard — into binary data. **The Memory Unit** holds this data, along with the program's instructions, until the CPU is ready for it. **The CPU** then takes over: the Control Unit fetches the next instruction (using the address held in the Program Counter), decodes it, and directs the ALU to carry it out — for example, performing an addition — with the result held temporarily in the Accumulator register. **The Output Unit** finally converts this binary result back into a form the user can understand, such as a number on screen.

Underlying every one of these exchanges is the System Bus: the Address Bus carries the memory address being accessed, the Data Bus carries the instruction or data itself, and the Control Bus carries the timing and Read/Write signals that keep every unit synchronised. Without the bus, the five units would simply be isolated boxes with no way to cooperate.

Worked example: consider a shopkeeper's billing system adding up a customer's bill. The Input Unit scans each item's barcode; the Memory Unit holds each item's price; the Control Unit fetches and decodes the “add to total” instruction for each item; the ALU performs the actual addition; the System Bus carries each price and address between memory and the CPU throughout; and finally the Output Unit displays the running total on screen and prints the final bill. Every single unit had to play its part — the CPU alone could not have produced a correct, printed bill.

In conclusion, the statement is entirely accurate: a block diagram's real value lies in showing organisation — the fixed roles of the Input Unit, CPU (Control Unit, ALU, and Registers), Memory Unit, Output Unit, and System Bus, and the precise, repeatable sequence in which they cooperate to turn a simple human action into a correct, useful result.

14 Exam Tips

- Memory trick for the five units: “I-C-M-O + Bus” — Input, CPU (Control Unit + ALU + Registers), Memory, Output, all connected by the System Bus.
- Memory trick for the buses: “Address = WHERE, Data = WHAT, Control = WHEN/HOW.”
- For “differentiate” questions (Input vs Output, ALU vs Control Unit, Primary vs Secondary memory), always answer using a table — it is faster to write and easier for the examiner to award marks clearly.
- Always draw the block diagram (Fig 3.1) whenever a question mentions “the basic parts of a computer,” “the CPU,” or “how a computer works” — diagrams carry dedicated marks in most university marking schemes.
- Common Mistake: Do not say the Control Unit “calculates” or “processes data” — that is the ALU's job. The Control Unit only directs and coordinates.

- Common Mistake: Do not confuse the address bus with the data bus — the address bus carries WHERE to look; the data bus carries the actual information.
- For 10 and 15 mark questions, structure your answer with a clear paragraph or table per component, exactly like the model answers in Section 13 — examiners reward clear structure over long, unstructured paragraphs.
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Lecture 3 — Block Diagram of a Computer — building directly on Lecture 1's Introduction to Computers.